📘 **AI/ML Internship – Day 1 Notes & Tasks**

🟢 **Day 1 Topic: Introduction to AI & ML + Git Basics**

📌 **Part 1: Introduction to AI & ML**

🔹 **What is Artificial Intelligence (AI)?**

Artificial Intelligence (AI) is the ability of a machine or computer system to perform tasks that normally require human intelligence.

✅ **Examples:**

- Voice assistants (Siri, Alexa)
- Recommendation systems (Netflix, YouTube)
- Self-driving cars
- Chatbots

🔹 **What is Machine Learning (ML)?**

Machine Learning is a subset of AI that allows machines to learn from data and improve performance without being explicitly programmed.

👉 Simple Meaning:
Instead of writing rules, we give data → machine learns patterns → makes decisions.

🔹 **Types of Machine Learning**

1. Supervised Learning

- Learns from labeled data
- Example: Predicting house price

2. Unsupervised Learning

- Finds patterns in unlabeled data
- Example: Customer segmentation

3. Reinforcement Learning

- Learns by trial and error

- Example: Game playing AI

🔹 **Real-Life Applications of AI/ML**

- Healthcare (disease prediction)
- Banking (fraud detection)
- E-commerce (product recommendations)
- Education (personalized learning)

📌 **Part 2: Introduction to Git**

🔹 **What is Git?**

Git is a version control system used to track changes in code and collaborate with others.

🔹 **Why Git is Important?**

- Keeps track of code history
- Helps in teamwork
- Easy to revert changes
- Widely used in industry

🔹 **Basic Git Commands**

```
git init          # Initialize repository
git add .         # Add files
git commit -m "message"  # Save changes
git status        # Check status
git log           # View history
```

🔹 **What is GitHub?**

GitHub is a platform to store and share your code using Git.

🎯 **Tasks for Day 1**

🧠 Task 1: Theory Questions

Answer the following in your own words:

1. What is Artificial Intelligence?
2. What is Machine Learning?
3. Difference between AI and ML
4. Explain types of Machine Learning with examples
5. List 5 real-life applications of AI

💻 Task 2: Git Installation & Setup

1. Install Git on your system
2. Verify installation:

```
git --version
```

3. Configure Git:

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

📌 Take screenshot and submit

💻 Task 3: Create Your First Repository

1. Create a folder named:

```
AI-ML-Internship
```

2. Initialize Git:

```
git init
```

3. Create a file:

```
day1.txt
```

4. Add and commit:

```
git add .
git commit -m "Day 1 completed"
```

📌 Take screenshot of terminal and submit

🌐 **Task 4: GitHub Account**

1. Create GitHub account
2. Create a new repository named:

AI-ML-Internship

3. Push your local project to GitHub

📌 Submit repository link

🚀 **Task 5: Research Task (Important)**

Search and write short notes on:

- AI vs ML vs Deep Learning
- Top 5 AI tools used in industry

🔥 **Task 6: Practical Thinking Task**

Think and write:

👉 "If you build an AI project, what problem will you solve and how?"

(At least 150 words)

🧩 **Task 7: Bonus Task (Extra Practice)**

- Install VS Code (if not installed)
- Create a simple text file and edit it
- Explore file saving and editing